REGULA-AI

Business Requirements Document (BRD) and Business Use Case

Team Byte Force

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AI-powered Product Compliance and Market Intelligence Platform for Indian Businesses

Purpose: A combined business requirements and use-case document for project evaluation.

PART A — BUSINESS REQUIREMENTS DOCUMENT (BRD)

1. Executive Summary

Regula-AI is a prototype decision-support platform for Indian product businesses, especially MSMEs, that brings preliminary BIS-related compliance research together with market intelligence, financial scenario planning, and growth recommendations. It helps a business organize the questions it should investigate before launching or expanding a product. It does not issue certification or replace official BIS decisions, legal advice, laboratory testing, or professional compliance services.

2. Business Problem

Manufacturers, importers, and product brands may find it difficult to identify which standards or Quality Control Orders could apply, what information they need to collect, and how compliance preparation could affect costs and launch planning. Compliance research and commercial planning are often handled separately. Businesses need a clearer preliminary workflow that connects compliance questions with market and financial considerations while identifying what remains unverified.

3. Project Objectives

• Provide preliminary product-compliance guidance and highlight requirements that need official verification.

• Generate a structured checklist of information and preparation tasks.

• Help users explore market demand factors, customer segments, competitors, and sales channels without inventing statistics.

• Estimate business scenarios such as revenue, operating profit, and break-even units from user-provided assumptions.

• Recommend practical next steps connecting compliance verification, market research, and business planning.

4. Target Users and Stakeholders

• MSME manufacturers preparing to introduce or expand a product.

• Importers and product brands researching Indian market-entry requirements.

• Founders and business teams comparing product-launch scenarios.

• Compliance consultants or business advisers using the platform as an initial research aid.

• Project evaluators and maintainers validating the prototype.

5. Scope

In scope

• Product name, description, and target-market input.

• AI-assisted analysis using the Gemini API.

• Curated BIS knowledge-base references for selected product categories, with evidence-status warnings.

• Preliminary compliance guidance and checklist generation.

• Market-intelligence and growth recommendations presented as research prompts or hypotheses.

• Financial scenario calculations based on user-entered assumptions.

• React/Vite frontend and FastAPI backend with documented API routes.

Out of scope / limitations

• Issuing BIS certification, approving products, or making binding legal determinations.

• Guaranteeing that a product does or does not require certification.

• Replacing testing, formal applications, qualified professionals, or official notifications.

• Guaranteeing demand, sales, profit, or growth outcomes.

• Claiming live market data or independently verified regulatory status without documented evidence.

6. Functional Requirements

ID	Requirement
FR-01 Product analysis	The system accepts a product name, product description, and target market and returns a structured preliminary analysis.

ID	Requirement
FR-02 Compliance guidance	The system describes potentially relevant compliance questions and distinguishes reference material from verified legal applicability.
FR-03 Checklist	The system provides preparation items or questions that a business should resolve before taking formal action.
FR-04 Market intelligence	The system outlines demand drivers, target customer segments, possible competitors, and sales channels; unsupported figures must not be fabricated.
FR-05 Financial scenario	The system calculates indicative financial metrics from user-provided assumptions and labels them as estimates.
FR-06 Growth recommendations	The system returns prioritized next steps that connect compliance, market research, and business planning.
FR-07 Health and API documentation	The backend exposes a health endpoint and interactive API documentation for testing.
FR-08 Safety and evidence labels	The system clearly states when source verification or product applicability is incomplete.

7. Non-Functional Requirements

• Usability: clear labels, structured sections, and plain-language explanations.

• Security: store API keys in an environment file and never commit secrets.

• Reliability: validate inputs and show useful errors when configuration or API calls fail.

• Transparency: label preliminary guidance, assumptions, and verification gaps.

• Maintainability: organize frontend, backend, and knowledge-base files separately.

• Performance: practical response times subject to network and AI-provider availability.

8. Solution Overview and Technology

The prototype uses a React + Vite frontend, a Python FastAPI backend, the Gemini API for AI-assisted analysis, and a curated BIS knowledge-base module. The frontend sends product or scenario information to the backend over HTTP/JSON. The backend validates requests, selects available reference context, calls the AI service where configured, and returns structured guidance. Financial scenario calculations use user-supplied assumptions.

9. API Overview

• GET / — basic backend status response.

• GET /api/health — health-check response.

• POST /api/analyze-product — product analysis, compliance research prompts, market intelligence, and recommended actions.

• POST /api/compliance-checklist — checklist generation.

• POST /api/financial-scenario — indicative financial scenario calculations.

Confirm exact request fields and response schemas against the current backend and Swagger UI at /docs before the final demonstration.

10. Assumptions and Dependencies

• Users provide reasonably accurate product descriptions and business assumptions.

• AI features require a valid Gemini API key and network access.

• The curated knowledge base covers selected categories; absence of an entry does not mean no requirement exists.

• Official standards, QCOs, applicability, dates, and market evidence must be checked independently.

• The project requires compatible Node.js/npm and Python packages.

11. Risks and Mitigations

Risk	Mitigation
Incorrect or outdated compliance guidance	Show evidence status and official-source verification warnings; do not present the AI output as a legal determination.
Hallucinated market numbers	Require source-backed figures or label assumptions; omit unsupported statistics.
API key exposure	Use .env locally, exclude it via .gitignore, and rotate a key immediately if exposed.
AI/API downtime	Show a clear error state and retain deterministic calculations where feasible.
Limited product coverage	State coverage limitations and expand the curated knowledge base only with reviewed references.

12. Success Criteria

• A user can submit a product description and receive a structured response.

• Outputs separate preliminary compliance guidance from verified facts and business assumptions.

• A user can generate a checklist and calculate a financial scenario.

• The application and API can be installed and run using the project documentation.

• The demo shows an end-to-end decision scenario without claiming certification or guaranteed commercial results.

13. Acceptance Criteria

The prototype is acceptable for evaluation when the frontend starts, the backend health endpoint responds, documented endpoints can be tested, AI functionality works with a valid key, and the repository excludes secrets. Compliance outputs must be labelled preliminary and financial outputs must be labelled estimates.

PART B — BUSINESS USE CASE

Use Case: MSME Evaluates Launching an Electric Kettle in India

Business scenario: An Indian MSME is considering manufacturing or importing an electric kettle. Before committing to inventory, packaging, testing, or distribution, the owner wants to understand which compliance questions to verify and whether the product launch appears commercially worth investigating.

Field	Description
Use Case ID	UC-01
Primary actor	MSME owner / product manager
Supporting system	Regula-AI web frontend, FastAPI backend, Gemini API, curated BIS knowledge-base reference
Trigger	The business is evaluating whether to proceed with an electric-kettle product launch.
Preconditions	The application is available; the user can provide a product description and, for AI analysis, the backend has a valid API key.
Main flow	1. The user enters “electric kettle,” a product description, and India as the target market. 2. Regula-AI checks whether its curated knowledge base contains a relevant reference. 3. The system generates preliminary compliance questions, evidence-status notes, and a checklist of information to verify. 4. The system outlines market demand factors, target segments, possible sales channels, and business considerations. 5. The user enters estimated selling price, unit cost, expected sales, fixed costs, and a possible one-time compliance budget in the financial scenario workflow. 6. Regula-AI calculates indicative revenue, operating profit, and break-even units. 7. The system presents prioritized next actions: verify official BIS/QCO applicability, confirm testing and documentation needs, validate demand using credible market sources, and review the financial assumptions.
Alternative / exception flows	If no product-specific reference exists, the system must say so and must not invent a standard or declare that certification is mandatory. If the Gemini key is missing or the service fails, the system should show a clear configuration/service error. If financial assumptions are incomplete, the user must be prompted to supply or correct them.
Postconditions	The user has a structured list of questions to verify and a preliminary commercial scenario to support further research; no certification decision has been made.
Expected value	Helps the business connect compliance preparation with launch cost, market research, and financial planning before making a larger commitment.
Success measure	The user can identify unresolved compliance questions, review the assumptions behind the financial estimate, and define concrete next steps.
Important limitation	Regula-AI is a preliminary decision-support tool. The user must verify official requirements through BIS/government sources and qualified professionals; market and financial outputs are not guarantees.

Business Decision Supported

The system does not decide whether the business is legally approved to sell the product. It helps the owner decide what to verify next and whether the product merits further compliance, market, and financial research.

Project Repository

https://github.com/Adithi030806/Regula-AI

Disclaimer: Regula-AI is a preliminary decision-support tool, not an official BIS service, legal opinion, certification authority, or guarantee of commercial outcomes.